Data Structures and Algorithms · 21CSC201J

Unit 1 · Session 5 · SLO 1 — Solved Worksheet

Name: DEEP SAND DEEP SAND | **Reg. No:** RA2511026010557

Unit â€“ 1 : Session â€“ 5 : SLO â€“ 1 | Dynamic Memory Allocation

I. Fill in the Blanks

1. run â€” dynamic memory is allocated at run time.
2. allocate â€” malloc() allocates raw memory.
3. deallocate (release) â€” free() releases memory.
4. initializes â€” calloc() allocates and zero-initializes memory.
5. resize â€” realloc() resizes previously allocated memory.
6. the size (in bytes) â€” sizeof() gives the size of a data type.
7. freed â€” a memory leak happens if memory is not freed.
8. void* (generic) â€” malloc() returns a void pointer.
9. malloc() â€” type casting is needed when using malloc in C (void* â†’ target type).
10. fails â€” NULL is returned if allocation fails.
11. 5 â€” malloc(5 * sizeof(int)) allocates memory for 5 integers.
12. dereference (*) / arrow â€” dynamic memory is accessed via the dereference or array operator.
13. heap â€” dynamic arrays use heap memory.
14. stack (static) â€” static arrays use stack/static memory.
15. use-after-free (dangling pointer access).

II. Code Snippets

1. int *arr = (int*) malloc(10 * sizeof(int));
2. float *f = (float*) calloc(5, sizeof(float));
3. arr = (int*) realloc(arr, 20 * sizeof(int));
4. free(arr); arr = NULL;
5. if (arr == NULL) { printf("Allocation failed"); exit(1); }